

Bruce Bainomugisha

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<https://annotation-portfolio.vercel.app/>

PROFESSIONAL SUMMARY

AI Data Specialist with Machine Learning background and hands-on experience in NLP and Computer Vision annotation projects. Certified in the **Google Advanced Data Analytics Professional Certificate** and trained in professional annotation workflows through **DataLens Africa**.

Experienced in NER annotation, instance segmentation, geospatial labeling, speech transcription, and AI response evaluation. Strong understanding of quality control processes, dataset bias, edge cases, and AI output judgment.

Detail-oriented, quality-focused, and experienced in working with annotation guidelines and SLA-driven environments.

CORE SKILLS

AI & Machine Learning

- Natural Language Processing (NLP)
- Named Entity Recognition (NER)
- Sentiment Analysis & Intent Classification
- Machine Learning Fundamentals
- Dataset Bias & Edge Case Handling
- AI Output Evaluation & Ranking

Computer Vision

- Object Detection (Bounding Boxes)
- Semantic & Instance Segmentation
- Polygon Annotation
- Keypoint Marking
- Video Object Tracking
- Image Classification & Localization

Geospatial & Remote Sensing

- GIS Fundamentals
- Satellite & Drone Imagery Annotation
- Land Cover Mapping
- Building Footprint Extraction
- Cadastral & Boundary Mapping
- Spectral Bands & Spatial Resolution

Speech & Audio Annotation

- Speech-to-Text Transcription
- Verbatim & Clean Read Transcription
- Speaker Diarization
- Acoustic Event Detection
- Environmental Sound Classification

AI Evaluation & Data Quality

- Hallucination Detection
- Instruction-following Evaluation
- Tone & Safety Analysis
- Rubric-based Scoring
- Gold Set & Benchmark Review
- Peer Review & Consensus Validation

Tools

- CVAT
- Label Studio
- QGIS
- Google Sheets / Microsoft Excel

CERTIFICATIONS

Google Advanced Data Analytics Professional Certificate – Coursera

- Data analysis, statistics, regression, ML foundations, data ethics

Essentials of Data Labeling & Annotation for AI Development – DataLens Africa**

- Professional annotation workflow
- SLA & client guideline compliance
- Quality Control (QC) processes
- Gold Sets & Benchmark Data
- Feedback loops & validation systems

PROJECT EXPERIENCE

Named Entity Recognition (NER) Project

Tool: Label Studio

- Annotated 40,000+ statements, with entities including Person, Organization, Location, Date, and Custom Labels
- Applied annotation guidelines with strict compliance to client rules
- Identified ambiguous and edge-case scenarios
- Participated in peer review and validation processes
- Improved dataset consistency through structured labeling decisions

Skills Applied: NLP, NER, guideline adherence, QC review, dataset bias awareness

Instance Segmentation Project

Tool: CVAT

- Performed polygon-based instance segmentation on 150+ images
- Maintained object ID consistency
- Applied quality control checks for annotation accuracy
- Handled complex overlapping objects and boundary precision

Skills Applied: Instance Segmentation, polygon annotation, object detection, edge-case handling

PROFESSIONAL ANNOTATION WORKFLOW KNOWLEDGE

Trained in structured annotation lifecycle:

- Onboarding & CAD Qualification

- Client Annotation Guideline Interpretation
- Main Project Work Under SLA
- Quality Control (Peer Review, Gold Sets, Consensus)
- Feedback & Iteration Loop

AI RESPONSE EVALUATION EXPERIENCE (SELF-TRAINED RESEARCH)

Developed expertise in:

- Evaluating AI-generated responses
- Detecting hallucinations & factual inconsistencies
- Ranking multiple responses with justification comments
- Writing ideal responses
- Rubric-based scoring
- Editing and humanizing AI-generated content
- Understanding quality vs spam signals
- Prompting for SEO intent

EDUCATION

Bachelors in Software Engineering
Makerere University Kampala
2019-2023